# Content-Based Movie Recommendation: Single-Item and Profile-Based Comparison

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## Abstract

This homework implements and compares single-item and profile-based content recommendation on the MovieLens 100K data using binary genre features and cosine similarity. A chronological leave-one-out experiment hides each eligible user's latest 5-star rating, builds both recommenders from earlier history, and evaluates their Top-5 rankings on 917 users. The single-item method achieved Hit Rate@5 of 1.418% (13 hits), compared with 1.091% (10 hits) for the profile method. The low accuracy and 19.56% mean list overlap show that aggregation changes ranking behavior but did not improve accuracy with these sparse genre-only vectors.

**Index Terms —** content-based recommendation, cosine similarity, user profile.

## 1. Introduction

### Problem statement

The application originally recommended movies by comparing one selected movie with every other movie using genre overlap. The homework extends this baseline in three stages: add cosine similarity, allow the user to select the recommendation method, and build a profile vector by averaging the genre vectors of three highly rated movies from a MovieLens user's history. The final experiment asks whether the Top-5 list from one active movie differs from the Top-5 list generated by the aggregated profile.

### Motivation

A single movie is a narrow expression of preference. A profile can retain features shared across several liked movies and suppress features that occur in only one, but aggregation can also make profiles generic and concentrate recommendations around common genres. An offline comparison is therefore needed rather than assuming that a profile is automatically more accurate.

### Contributions

- Implement Jaccard and cosine similarity over correctly parsed MovieLens genre flags, with a user-selectable method in the browser interface.
- Implement a profile-based cosine recommender that averages three movie vectors, excludes watched movies, sorts deterministically, and displays a score for every numbered recommendation.
- Compare single-active-item and profile-based Top-5 rankings with a chronological 5-star holdout protocol over 917 eligible users, audit popularity exposure and profile collapse, preserve the measured results as reproducible JSON artifacts, and expose the per-user comparison in the browser interface.

## 2. Related Work

The MovieLens 100K dataset represents preferences as user, item, rating, and timestamp tuples; the local copy contains 100,000 ratings from 943 users for 1,682 movies [1]. Movie records provide binary genre indicators, which makes the dataset suitable for a small, interpretable content-based recommender.

Cosine similarity is the length-normalized dot product of two vectors [2]. With binary genre vectors, the dot product counts shared genres, while the denominator adjusts for how many genres each movie has. Content-based recommenders commonly represent item descriptions and a profile of user interests, then compare candidate items with that profile [3].

Sarwar et al. introduced item-based collaborative filtering that learns item similarity from the user-rating matrix [4]. The "item-to-item" arm in this homework is different: it compares the content features of one active movie with candidate content features and does not learn similarity from co-rating behavior. This distinction prevents the experiment from claiming evidence about collaborative filtering.

Ranked retrieval evaluation considers the top *k* positions and uses discounted gain to give more credit when a relevant item appears earlier [5]. Because this experiment has one held-out relevant movie per user, Hit Rate@5 measures whether it appears, MRR@5 rewards an earlier reciprocal rank, and NDCG@5 applies logarithmic rank discounting.

### Alternatives considered

- **Rating-weighted profile:** weight genre vectors by rating rather than selecting only three movies. This would use more preference strength but changes the requested three-title averaging scenario.
- **All-history centroid:** average every positively rated movie. This may be more stable, but can blur distinct tastes and makes the requested three-movie example less transparent.
- **Collaborative item similarity:** learn similarities from rating co-occurrence. This is a valid recommender, but it is a different method from the requested genre-vector cosine implementation.

## 3. Method

### Application logic

Each movie is represented by an 18-dimensional binary vector covering `Action` through `Western`. The separate MovieLens unknown flag is not treated as `Action`; named genres are read from fields 6 through 23 of `u.item`. For movie vectors `x` and `y`, cosine similarity is the normalized dot product:

$$s_cos(x, y) = \frac{x \cdot y}{\|x\|_2 \times \|y\|_2}$$

For the corresponding sets of active genres `Gx` and `Gy`, Jaccard similarity is:

$$s_J(G_x, G_y) = \frac{|G_x \cap G_y|}{|G_x \cup G_y|}$$

The single-item modes compare a selected movie with all other movies using either cosine or Jaccard similarity. The profile mode chooses the user's `m = 3` highest-rated movies; newer timestamps and then smaller movie IDs break ties. Their vectors are averaged component-wise:

$$p = \frac{1}{m}\sum_{i=1}^{m} x_i, \qquad m = 3$$

and an unseen candidate `x_j` receives the normalized profile score:

$$s_profile(x_j) = \frac{p \cdot x_j}{\|p\|_2 \times \|x_j\|_2}$$

The implementation returns 0 when a cosine denominator or Jaccard union is zero, avoiding division by zero.

A fourth experiment mode applies the chronological holdout protocol to a selected user and displays both Top-5 lists side by side, together with the hidden positive, each method's target similarity and positive-similarity rank, Hit@5 outcomes, and overlap. The UI explicitly separates the actual 5-star user rating from cosine similarity, which ranges from 0 to 1 and is not a predicted star rating. All UI results use an ordered list, descending scores, alphabetical tie-breaking, and a displayed score rounded to three decimals.

#### Offline comparison protocol

The comparison script uses a chronological leave-one-out design:

1. Sort each user's ratings by timestamp and movie ID.
2. Hold out the latest 5-star rating that has at least three earlier interactions.
3. Build all model inputs only from ratings before the held-out interaction.
4. Select the three highest-rated training movies, breaking ties by newest timestamp and movie ID.
5. Use the first selected movie as the single active item and the mean of all three genre vectors as the profile.
6. Rank the same candidate pool—all catalog movies absent from the training history—with cosine similarity and retain Top-5.

This yields 917 eligible users. Twenty-six users are not evaluated because the required chronological 5-star split is unavailable. Interactions after the chosen target do not train either method; the candidate pool is the catalog minus training items.

#### Metrics

For each user, the held-out positive movie is the only relevant item. Let `N` be the number of evaluated users, `r_u` the target rank for user `u`, `I_u = 1` when `r_u ≤ 5` and `I_u = 0` otherwise, and `R_u` the user's Top-5 set. The metrics are:

$$HR@5 = \frac{1}{N}\sum_{u=1}^{N} I_u, \qquad MRR@5 = \frac{1}{N}\sum_{u=1}^{N} \frac{I_u}{r_u}$$

$$NDCG@5 = \frac{1}{N}\sum_{u=1}^{N} \frac{I_u}{\log_2(r_u + 1)}, \qquad Coverage@5 = \frac{|\bigcup_{u=1}^{N} R_u|}{1682}$$

Because there is one relevant target per user, a miss contributes zero to MRR@5 and NDCG@5. List agreement is measured by $|R_u^{item} \cap R_u^{profile}|$ and by the fraction of users with exactly identical ordered lists.

#### Environment and reproducibility

- Dataset: local MovieLens 100K files `u.data` and `u.item`.
- Runtime: Node.js v23.7.0, standard library only.
- Randomness: none; all sorting and ties are deterministic.
- Commands from the `week2` directory: `compare_item_profile.js`, `analyze_mainstream_bias.js`, `analyze_profile_collapse.js`, and `run_report_checks.js` inside `artifacts/` (run each with Node.js).
- Saved results inside `artifacts/`: `item_profile_results.json`, `mainstream_bias_results.json`, `profile_collapse_results.json`, and `report_check_results.json`.

## 4. Experiments

#### Implementation sanity experiments

The selectable single-item methods were checked with *Toy Story (1995)*. Both ranked *Aladdin and the King of Thieves (1996)* first with score 1.000; cosine ranked *Aladdin (1992)* second with 0.866025, while Jaccard assigned the same movie 0.750000, confirming that the UI executes distinct formulas. The ordinary full-history profile mode was checked for User 1 using *Truth About Cats & Dogs, The (1996)*, *Delicatessen (1991)*, and *Kolya (1996)*, all rated 5. Its five unseen Comedy recommendations each scored 0.904534, matching the transparent profile calculation in the verification section. The remaining experiments use the chronological held-out protocol: aggregate item/profile evaluation, a per-user comparison, popularity exposure, and profile-collapse diagnostics.

**Aggregate results**

**Table 1. Chronological Top-5 evaluation over 917 users**

| Metric | Single active item | Aggregated profile |
|---|---|---|
| Hits@5 | 13 | 10 |
| Hit Rate@5 | 0.014177 (1.418%) | 0.010905 (1.091%) |
| MRR@5 | 0.008270 | 0.005071 |
| NDCG@5 | 0.009765 | 0.006506 |
| Distinct recommended movies | 505 | 463 |
| Catalog coverage | 0.300238 (30.02%) | 0.275268 (27.53%) |

The single-item method found 13 held-out 5-star movies, while the profile method found 10. It also had higher MRR@5 and NDCG@5, so this run provides no accuracy improvement from three-movie profile aggregation. These differences are descriptive and no uncertainty test was performed. The single-item method covered 42 more catalog movies and 2.50 percentage points more of the catalog.

**Recommendation-list agreement**

**Table 2. Agreement between the two Top-5 lists**

| Measure | Result |
|---|---|
| Mean shared movies per user | 0.978 of 5 |
| Mean overlap fraction | 0.195638 (19.56%) |
| Users with identical ordered Top-5 | 102 of 917 (11.12%) |

On average, fewer than one of five recommended movies was shared, and only about one user in nine received exactly the same ordered list. Aggregating multiple movies therefore changes the ranking substantially, but in this run it also produced fewer hits than the single-item method.

**User 1 case study**

For User 1, the held-out target was *Delicatessen (1991)*, rated 5. The training profile used *Truth About Cats & Dogs, The (1996)*, *Kolya (1996)*, and *Crumb (1994)*, all rated 5. The single-item arm used *Truth About Cats & Dogs, The (1996)* as its active movie.

**Table 3. User 1 Top-5 recommendations**

| Rank | Single active item | Score | Aggregated profile | Score |
|---|---|---|---|---|
| 1 | Addicted to Love (1997) | 1.000000 | Addicted to Love (1997) | 0.866025 |
| 2 | Annie Hall (1977) | 1.000000 | Annie Hall (1977) | 0.866025 |
| 3 | Barcelona (1994) | 1.000000 | Barcelona (1994) | 0.866025 |
| 4 | Beautician and the Beast, The (1997) | 1.000000 | Beautician and the Beast, The (1997) | 0.866025 |
| 5 | Benny & Joon (1993) | 1.000000 | Benny & Joon (1993) | 0.866025 |

The lists are identical, but neither contains the held-out target. The single-item method assigns *Delicatessen* cosine similarity `0.500` and positive-similarity rank 340; the profile assigns `0.577350` and rank 297. Thus, the profile evaluates the target as somewhat closer without placing it in Top-5. Multiple score ties show a limitation of only 18 binary genre features: many movies become indistinguishable and alphabetical ordering decides their relative positions.

**Mainstream-domination check**

Movie popularity was measured after recommendation using the number of ratings in the complete `u.data` file; it was not supplied to either model. A high-visibility movie was defined before inspecting the cases as one in the catalog's top rating-count quartile, which begins at 80 ratings. The User 1 lists are identical, so Table 4 records the diagnostic metadata once for each shared recommendation.

**Table 4. Popularity metadata for User 1's shared Top-5**

| Rank | Movie | Rating count | Release year | Top popularity quartile |
|---|---|---|---|---|
| 1 | Addicted to Love (1997) | 54 | 1997 | No |
| 2 | Annie Hall (1977) | 180 | 1977 | Yes |
| 3 | Barcelona (1994) | 53 | 1994 | No |
| 4 | Beautician and the Beast, The (1997) | 86 | 1997 | Yes |
| 5 | Benny & Joon (1993) | 102 | 1993 | Yes |

Thus, three rather than all five User 1 recommendations were high-visibility by the declared rule. Across all 917 users, the 4,585 recommendation occurrences from the single-item method had mean/median rating counts of 71.2/40 and a 33.0% high-visibility share. The profile method had mean/median counts of 88.7/57 and a 42.6% high-visibility share, compared with 59.5/27 and 25.1% for the catalog. This is evidence of popularity enrichment in exposure, particularly for the profile method, but not proof that the algorithms measure popularity: rating count is absent from the score, and the correlations between rating count and catalog-wide recommendation frequency were only 0.040 and 0.105.

The proposed dense-vector mechanism is also not generally valid. With one shared genre, cosine is $1/\sqrt{|Q| \times |C|}$, so adding unmatched candidate genres increases the denominator and lowers the score. Empirically, the correlation between genre count and cosine score over every catalog movie for every eligible-user model was 0.003 for single-item and 0.037 for profile-based scoring, both close to zero.

Recommended occurrences nevertheless averaged 2.10 and 2.73 genres for the two methods versus 1.72 in the catalog, and genre count correlated with recommendation frequency at 0.120 and 0.336. Profile recommendation therefore shows multi-genre exposure enrichment, but not a general monotonic score reward for vector density. As a declared niche-preference proxy, User 707 was selected because their three profile inputs had the lowest mean rating count among eligible users (15, 8, and 4 ratings; mean 9). Neither method placed any top-quartile-popularity movie in that user's Top-5: the single-item counts were 5, 26, 8, 3, and 1, while the profile counts were 39, 3, 26, 29, and 8. The strong claim that unusual preferences necessarily produce five mainstream recommendations is therefore rejected for this experiment, while the weaker aggregate enrichment finding remains.

#### Profile-averaging collapse check

The suggested comparison with each input movie's self-similarity needs care: self-similarity is always 1, so any non-identical average usually scores below it. Indeed, all three profile-to-input similarities were below 1 for 902 of 917 users (98.4%), making that condition too permissive to establish washout. The stronger diagnostic marked a recommendation as *centroid-preferred* when its profile score exceeded its similarity to every one of the three individual input movies. Input coherence was the mean of the three pairwise cosine similarities; lower values mean more diverse inputs.

**Table 5. Profile behavior by input-coherence quartile**

| Metric | Lowest coherence | Highest coherence |
|---|---|---|
| Users | 230 | 230 |
| Mean pairwise input similarity | 0.075 | 0.648 |
| Mean profile-to-input similarity | 0.608 | 0.871 |
| Centroid-preferred Top-5 occurrences | 47.1% | 16.3% |
| Users with an input absent from all Top-5 genres | 57 (24.8%) | 0 (0.0%) |
| Profile Hit Rate@5 | 3/230 (1.30%) | 5/230 (2.17%) |

The results support a bounded collapse effect: diverse profiles generate substantially more hybrid centroid matches and more often omit one input taste entirely. The Hit Rate difference is descriptive only because it represents three versus five hits and no uncertainty test was performed. User 3 was selected by the declared lowest mean pairwise-similarity rule; all three input pairs had cosine 0. Its profile inputs were *Mother* (Comedy; profile similarity 0.408), *Conspiracy Theory* (Action/Mystery/Romance/Thriller; 0.816), and *Boogie Nights* (Drama; 0.408). Every one of its five recommendations was centroid-preferred: profile score versus best individual-input score was 0.913/0.894 for *Diva*, 0.816/0.750 for *Charade*, and 0.707/0.577 for each of *American President, The*, *Apollo 13*, and *Arsenic and Old Lace*. This is direct evidence of hybridization in genre space, but the data cannot determine whether a person perceives those movies as "generic." User 1 shows the other failure mode: all five profile recommendations match Comedy/Romance, while the Documentary input *Crumb* has zero similarity to every Top-5 item and is completely unrepresented.

#### Verification

The experiment was executed twice. After excluding the timestamp field `generatedAt`, the JSON outputs were identical. Independent arithmetic checks recomputed the hit rates from `13/917` and `10/917`, the overlap fraction from `0.978190/5`, and both coverage fractions from `505/1682` and `463/1682`.

A transparent profile-score check is available for User 1. The three selected movies produce nonzero profile weights `Comedy = 2/3`, `Romance = 1/3`, and `Documentary = 1/3`. A Comedy-and-Romance candidate therefore scores

$$score = \frac{(2/3) + (1/3)}{\sqrt{(4/9) + (1/9) + (1/9)} \times \sqrt{2}} = \frac{\sqrt{3}}{2} = 0.866025\ldots$$

which matches the stored score and the UI value `0.866` after rounding.

For the held-out target *Delicatessen*, the active movie and target share one of two genres each, giving single-item cosine $1/(\sqrt{2} \times \sqrt{2}) = 0.5$. The profile shares only its $Comedy = 2/3$ component with the target's Comedy and Sci-Fi vector, giving cosine $1/\sqrt{3} = 0.577350\ldots$. The saved full positive-similarity rankings place these scores at positions 340 and 297, respectively.

The browser experiment mode was checked for two users. User 1 reproduced the saved five-star case study exactly, including two misses and full overlap. User 8 exercised the hit state: both lists placed the held-out *Star Trek III: The Search for Spock (1984)* at rank 2, with displayed scores `1.000` and `0.927`. The successful rows were visibly marked as the held-out target.

## 5. Discussion

#### Interpretation

Profile aggregation usually changed which movies were recommended, but it reduced every measured accuracy metric in the five-star holdout run. The absolute values are extremely low, so this is evidence only for the current sparse genre-only setup rather than a universal advantage for single-item recommendation. Genre-only content also cannot distinguish movies with identical genre combinations, which creates large score ties and lets title order affect Top-5 membership.

#### Required checks

- **Overlap check:** Across all 917 comparison users, both Top-5 lists had an empty intersection with the visible watched/training history—0 violations among 9,170 recommendation occurrences—and ordinary profile mode likewise had 0 violations among 4,715 occurrences for all 943 users; the deliberately hidden target remains eligible by definition.
- **Distribution check:** User 1's shared Top-5 had rating counts 54 (*Addicted to Love*), 180 (*Annie Hall*), 53 (*Barcelona*), 86 (*Beautician and the Beast, The*), and 102 (*Benny & Joon*), so only 3/5 met the declared mainstream threshold of 80 rather than all five.
- **Formula check:** The §3 formula and both implementations normalize by vector magnitudes—an example with raw dot product 1 and magnitudes $\sqrt{2}$ and $\sqrt{2}$ returned cosine 0.500 for both binary-item and numeric-profile functions, rather than the unnormalized value 1.

### Failure cases and fixes

1. **Shifted genre labels.** Symptom: `unknown` appeared as `Action`, later genres shifted, and `Western` disappeared. Cause: named genres began at field 5 although field 5 is MovieLens's separate `unknown` flag. Fix: read named genres from fields 6–23; *Toy Story* and *GoldenEye* then produced their expected genres.
2. **Different browser and offline ranks.** Symptom: target ranks differed for the same User 1 case. Cause: the two paths used different character decoding, locale-sensitive `localeCompare`, and different genre-flag comparisons. Fix: use ISO-8859-1 `TextDecoder`, numeric genre flags, and explicit code-point title ordering in both paths; both now return ranks 340 and 297.
3. **Stale browser resources.** Symptom: the browser could display an older script or stylesheet after local edits. Cause: cached resources were reused while filenames stayed unchanged. Fix: add matching version query parameters to `style.css`, `data.js`, and `script.js`; a fresh load retrieved the current resources.
4. **Unauditable ranking output.** Symptom: ranking and scores were not consistently visible. Cause: results were not uniformly represented as a numbered, scored list. Fix: render ordered lists sorted by descending score and show a score for every recommendation; UI and automated checks confirmed the order and values.

Movie titles such as *Client, The (1994)* are preserved from MovieLens's article-at-the-end naming convention; this is source-data formatting, not a word-order calculation bug.

### Limitations

- One held-out positive per user is a strict evaluation and treats every unobserved catalog movie as non-relevant, although many were never rated.
- Timestamps can be tied; movie ID is only a deterministic tie-breaker and does not reveal the true within-second order.
- The profile is an unweighted mean of three movies and ignores rating magnitude after selection, negative preferences, actors, directors, plots, and collaborative evidence.
- Alphabetical tie-breaking is reproducible but not preference-aware.
- Full-dataset rating count is only a post-hoc visibility proxy; it mixes exposure and preference and cannot establish a causal popularity effect.
- This experiment reports descriptive metrics only; it does not include confidence intervals or paired significance tests.

I was suprised that the cosine item-2-item method gives better results for recommendation of films then the user profile.

## 6. AI Usage Disclosure

### AI tools used

- OpenAI Codex, used for implementation support, experiment design and execution, browser-oriented checks, arithmetic verification, and report drafting.

### How AI was used

The AI inspected the starter files, corrected parsing, encoding, ordering, and display issues, implemented cosine and profile-based recommendation logic, added the UI method selector and interactive comparison, created deterministic item/profile, popularity, profile-collapse, and report-check scripts, executed and cross-checked the experiments, verified external references, and drafted this report. The evidence ledger distinguishes AI checks from student checks.

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